

Application Deployment on Cloud (AWS)

Step-1: Project Setup

Please set up the project folder on your machine, ideally at the desktop location, and use the suggested project name **fullstack-s104** or **fullstack-s201** which MUST contains the following folders & file

project_front (folder)

project_back (folder)

pipelinecode (txt file) (refer to email)

Note: Keep all these in one folder either in fullstack-s104 or fullstack-s201

Step-2: EC2 (Elastic Compute Cloud) Instance Creation

a. Instance Creation

Step-1: Search for EC2 in AWS search window

Click on EC2

Step-2: Click on Launch Instance

Name and tags: machine1

Application and OS images: Linux

Instance type: t3.medium or t3.small

Key pair(login): default

Network Settings:

- Firewall (security groups)
- Choose existing firewall

Configuration Storage: 20

Click on Launch Instance

Step-3: Add Security Group to Instance Created

- Go to left side panel
- Go to Networks & Security

- Security Groups
- Click on Security Group ID
- Click on Inbound rules
 - Add Rule
 - Type: all traffic
 - Source: Anywhere-IPV4
- Click on save rule

Step-4: Add Elastic IP to the instance created

- Go to left side panel
- Go to Networks & Security
- Click on Elastic IP's
- Click Associate Elastic IP address
- Choose Instance (machine1)
- Click on Associate

Step-3: RDS (Database Creation)

Step-1: Create database

- Database creation method: standard create
- Engine options: mysql
- Templates: Free-tier
- Settings:
 - DB Instance Identifier: database-1
 - Master username: admin
 - Master password: adminadmin
- Public access: Yes
- Additional Configuration:
 - initial database name: onetwo
- Click on create database

Step-4: Installation of Tomcat & Jenkins in AWS

- Refer to previous email which consists of PDF document for understanding Installation of Tomcat & Jenkins in AWS

Step-5: Access (Manage) Jenkins – Install Plugins & Tools

Plugins: git push

pipeline: multibranch with defaults

maven integration

nodejs

Tools:

JDK_HOME

MAVEN_HOME

NODE_HOME

(Refer to previous email which consists of WORD document for understanding Installation of Plugins & Tools in Jenkins)

Step-6: Push the project from local to GIT repository

- Refer to earlier concepts to understand how to push the project from local to remote repository

Step-7: Database properties (application. properties) in crud_back

Do the changes in github remote repository

database-source: localhost must be replaced with db-endpoint URL

username: admin

password: adminadmin

Step-8: Open App.jsx file in crud_front

Do the changes in github remote repository

App.jsx:

BASE URL:

Replace local host with instance IP address

Change port number

Step-9: Create a Job in Jenkins

Click on Item

- provide
 - choose pipeline
- click on save

Description: This is our full stack project

Poll SCM: * * * * *

Pipeline Code:

1. Refer to email and copy the pipeline code here & also make sure, it is available in remote repository too.
2. Make necessary changes in pipeline code
 - Tomcat URL: your IP address with port number
 - github remote repository of yours